

Joseph Zou

(650) 483-8310 | josephzou05@gmail.com | linkedin.com/in/josephzou05 | josephzou05.github.io

EDUCATION

Purdue University

West Lafayette, IN

Bachelor of Science in Computer Science - Data Science, GPA: 3.91/4.0

Aug 2023 – May 2026

- **Relevant Coursework:** Artificial Intelligence, Data Mining & Machine Learning, Statistical Programming & Data Management, Data Visualization, Data Structures & Algorithms, Large Scale Data Analytics

EXPERIENCE

Gen AI Platform Engineer Intern (Incoming) @ Lenovo

Jun 2026 – Jul 2026

- Incoming Gen AI Platform Engineer Intern contributing to scalable ETL pipelines and AI platform infrastructure for production ML systems
- Building Python and SQL data workflows to support AI-driven AR/VR applications and system integration
- Supporting development of REST-based microservices and data workflows for reliable model integration

Data Science Researcher @ John Deere

Aug 2025 – Dec 2025

- Designed and trained tree-based machine learning models (Decision Trees, Random Forests, XGBoost) for 3-month part-level demand forecasting across multi-year time series data
- Implemented recursive forecasting to generate realistic, sequential predictions without data leakage
- Engineered lag-based and temporal features to capture short-term demand patterns across 67-month time series
- Evaluated model performance using RMSE to guide improvements in forecast accuracy and business relevance

Data Science Intern @ BASF

Aug 2024 – May 2025

- Analyzed 10 years of California miticide application data to identify seasonal and regional usage patterns
- Estimated reapplication probabilities conditioned on spring active ingredient choice
- Conducted cost-benefit simulations to compare long-term outcomes of treatment strategies
- Performed geospatial county-level segmentation to inform region-specific product recommendations

PROJECTS

Interpretable Generative Planning for Autonomous Driving

Jan 2026 – Present

- Collaborating in a capstone research team at Purdue University, advised by Prof. Ziran Wang and Dr. Jiaru Zhang, on diffusion-based autonomous driving
- Implementing bottleneck latent representations inspired by SODA to improve interpretability of generative trajectory planning models
- Adding auxiliary objectives to encourage semantically meaningful latent representations of driving behavior
- Evaluating planning performance and representation quality on large-scale autonomous driving datasets using the NAVSIM simulation framework

NBA Prop Prediction | *Python, Pandas, scikit-learn*

Dec 2025 – Present

- Built an end-to-end pipeline to predict NBA player points using historical performance and game context
- Engineered rolling and opponent-specific features to capture recent form, minutes trends, and matchup effects
- Trained and evaluated linear regression and gradient-boosted tree models using RMSE and MAE on held-out data
- Developed model interpretability and evaluation visualizations, including feature importance and prediction calibration

Body Image & Gym Habits ML Analysis | *Python, Pandas, NumPy, Matplotlib*

May 2025 – Aug 2025

- Built an end-to-end machine learning pipeline on behavioral survey data, from preprocessing to evaluation
- Trained and compared logistic regression, decision tree, and random forest models to predict body dissatisfaction risk based on behavioral and demographic survey variables
- Evaluated model performance using ROC-AUC and recall across linear and tree-based classifiers
- Applied K-means and hierarchical clustering to uncover gym-goer archetypes and quantify large disparities in body dissatisfaction risk, including one cluster with ~ 68% at-risk prevalence

TECHNICAL SKILLS

Languages: Python, R, SQL, SAS, Java, C/C++ **Libraries:** pandas, NumPy, scikit-learn, matplotlib, ggplot2, NLTK